Multi-view Shape Reconstruction Based on M-array Image Decoding with Adaptive Local Window

Chen Luo, Zhijie Mo and Yijun Zhou
School of Mechanical Engineering
Southeast University
Nanjing, Jiangsu Province, China, 21189
{chenluo, mozhijie, zhouyj}@seu.edu.cn

***Abstract* - Three-dimensional (3D) shape reconstruction using coded structured light is considered one of the most reliable techniques to recover object surfaces. This paper aims to address problem of finding correspondences in a structured-light 3D shape reconstruction system. Existing technique (under M-array Image Decoding) applies exhaustive brute force algorithm, which suffers from slow processing and tends to reject valid symbols during its decoding. We propose a novel method based on local dynamic window to effectively address the key correspondence problem in the presence of occlusion. The dynamic window can adapt by taking into account local information in challenging areas of edges, corners and occlusion-induced discontinuity to improve decoding accuracy. Compared with existing approaches, the proposed technique enables imaging and reconstructing object shapes with conventional hardware at higher resolution.**

Index Terms - Color recognition, feature extraction, multi-view reconstruction, pseudo-random coding.

I. INTRODUCTION

Image an object and then reconstruct three-dimensional geometry of that object is an important research topic across many fields with applications in robotic vision, defense, remote sensing, medical imaging, manufactured part inspection, and autonomous vehicles [1]. Stereo vision has been widely used to extract 3D information from multiple 2D views of a scene. Multiple images are acquired from different points of view on the same object and 3D information can be obtained by examining the relative positions of object in the images [2]. Key step of that involves finding corresponding 2D image pixels (or points) for 3D world point on the object.

Given a pixel in an image, finding the corresponding pixels in other images is a well-known challenge in computer vision. Surface textures of target object have considerable impact on 3D reconstruction. Weak texture or textureless object has difficult to be reconstructed [3] as only limited number of pixels can be successfully paired. For complex texture (or thin structures), it tends to have the problem of points mismatching. Although the issue can be somehow alleviated by adding geometric constraints, the pixel matching still stands as a big problem [4] in many applications.

Methods based on coded structured light have been proposed to address the correspondence issue by giving specific code-words to every unitary position in the image. As such, coded structured light is considered one of the most reliable techniques for recovering the surfaces of objects. This technique is based on projecting a light pattern and imaging the illuminated scene from one or more points of view. Since the pattern is coded, correspondences between image point and point of the projected pattern can be found through mapping.

Available coded structured light methods include Binary codes [5], N-ary codes [6], Gray code [7] and De Bruijn sequence [8]. However, M-array (constructed by folding a pseudorandom sequence) turns out popular due to interesting mathematical properties of those matrices. Existing method applies brute-force algorithm to conduct exhaustive search and pairing based on a fixed rectangular window. However, applying this method has two issues for concern. First, the correspondence-finding step is prohibitively computationally demanding in many application cases, in terms of both memory requirements and processing cycles. Second, the method rejects considerable amount of valid point-pairs resulting in low pairing rate, thus preventing its widespread use.

In this paper, we aim to overcome the correspondence challenge by proposing an adaptive local window method. Under the proposal, the window is constructed according to the arrangement direction of the 4-nearest feature points based on measured distance. Then further search is conducted to identify 4 nearest points at each vertex of the window in an effort to arrange 9 points in the local window. The correspondence of feature points between the captured image and the projected pattern can be done through these local windows. The dynamic window can adjust itself to taking into account local information to tackle challenging areas along edges, corners, and occlusion-induced discontinuity. The proposed method, together with developed robust algorithm, considerably improves the decoding speed (showing the potential for real-time, or near real-time application) and decoding rate (meaning higher resolution for subsequent reconstructed shapes).

This paper is structured as follows. Section II provides related work on coded structured light based methods. Section III presents detailed methodology on M-array generation, dynamic window construction and decoding algorithm. A case example is presented in Section IV to illustrate the proposed method and verify its effectiveness. We summarize the study and provide suggestions for future research in Section V.

This work was partially supported by the National Science Foundation of China under grant No. 51105075 and No. 51575107.

II. Related Work

Over the past decades, various methods have been proposed by researchers on codification of projected patterns, which is used to improve matching (or mapping) accuracy with additional encoded information. Posdamer and Altschuler [9] pioneered the research by proposing the projection of a sequence of m patterns to encode 2^m stripes using a plain binary code. Subsequently, Inokuchi *et al*. [6] improved the codification scheme by introducing Gray code (instead of plain binary) which has consecutive code words with a Hamming distance of one and more robustness against noise. Later on, Caspi *et al*. [10] proposed a multilevel Gray code based on color. The extension of Gray code is based on an alphabet of n symbols, where every symbol is associated to a certain RGB color.

Patterns based on Gray code, as well as binary and n-ary codes, have the advantage that the pixel codification is made punctually with no extra spatial neighborhood checking required in the codification. However, the discrete nature of such patterns limits the range resolution. To address that, researchers [11, 12, 13] tried to integrate Gray Code Methods (GCM) with Phase Shift Methods (PSM), designed by Bergmann [14], to exploit higher spatial resolution with unambiguity and robust codification to improve 3D reconstruction accuracy. However, number of projecting patterns increased considerably under this hybrid approach.

Codification methods based on grey levels and color were proposed by researchers [15, 16, 17, 18, 19] so that every pixel can be labeled by the information represented on pattern. In theory, a high resolution of 3D information can be obtained. However, the hybrid approach is quite sensitive to noise. Also, the imaged colors depend not only on the projected colors, but also on the intrinsic color of the measuring surface. Therefore, direct codification is usually constrained to neutral color objects or pale objects in applications.

Vuylsteke and Oosterlinck [20] made a notable breakthrough to present a binary encoded pattern by means of De Bruijn sequences. A total number of 63 columns were encoded in a unique pattern, which is also suitable for moving scenes.

First presented by Etzion [21], M-arrays (perfect maps) were random arrays of dimensions $r \times v$ in which a sub-matrix of dimensions $s \times t$ appeared only once in the pattern. M-arrays represented in a two-dimensional space what De Bruijn patterns were in a one-dimensional space [21]. The code word that labeled a certain point of the pattern was obtained from a neighborhood of the points around it [22]. Morita *et al*. [23] proposed a two projection based technique where an encoded matrix of black dots on a white background was projected, while in the second projection some black dots were removed according to a binary-encoded M-array. Morano *et al*. [24] proposed a brute force algorithm to generate the pattern. An iterative algorithm adding one new code word and checking it against all the previous ones was performed. If all the distances between values were at least equal to the specified minimum Hamming distance, the new word was accepted and the next iteration was followed, until the pattern was created.

In view of existing work, this paper proposes a dynamic window approach in M-array image decoding. This method requires much smaller computational and memory resources than Morano's method [24] and images objects at unprecedented resolution. The proposed methodology is presented in below section.

III. Methodology

A. Pattern Based on M-array Codification

Let **M** be a matrix of dimensions $r \times v$ where each element is taken from an alphabet of k symbols {0, 1, …, k-1}. If **M** has the window property, i.e. each different submatrix of **M** of dimensions $n \times m$ appears exactly once, then **M** is a *perfect map*. If **M** contains all submatrices of $n \times m$ except the one filled by 0's, then **M** is called an *M-array* or *pseudorandom array* [25]. This type of array has been widely used in pattern codification because the window property allows every different submatrix to be associated with an absolute position in the array.

M-array can be constructed by folding a pseudorandom sequence [26]. To construct a pseudo-random sequence of length $n = q^m$-1, one needs a primitive polynomial $h(x)$ of degree m:

$$h(x) = x^m + k_{m-1}x^{m-1} + \mathrm{K} + k_2 x^2 + k_1 x + k_0 \tag{1}$$

where $k_i \in GF(q)$ [1] and $k_0 \neq 0$. The feedback registered sequence can be expressed as follows:

$$\begin{aligned} a_{i+m} = h(a_{i+m-1}, a_{i+m-2}, \mathrm{K}, a_i) = -k_{m-1}a_{i+m-1} \\ -k_{m-2}a_{i+m-2} - \mathrm{K} - k_1 a_{i+1} - k_0 a_i \end{aligned} \tag{2}$$

Here, $a_{i+m} = a_{i+m} \bmod q$ for $\forall a_{i+m} \notin GF(q)$.

If $m = k_1 \times k_2$, where the related window property is $k_1 \times k_2$, M-array is binary matrix of $n_1 \times n_2$, where $n_1 = q^{k_1} - 1$, $n_2 = n / n_1$, $k_1 > 1$ and $k_2 > 1$.

For 3D shape reconstruction, we can select an alphabet of 3 symbols {0, 1, 2} and window property 3 × 3. So q=3 and m=9. The corresponding primitive polynomial is $h(x) = x^9 + x^7 + x^5 + 1$ and

$$a_{i+9} = -a_{i+7} - a_{i+5} - a_i \quad (i = 1, 2, \ldots, 3^9 - 10) \tag{3}$$

Set initial value as 100000000, the pseudorandom sequence generated according to (3) is as follows:

$a_1 a_2 a_3 a_4 a_5 \mathrm{L}\ a_{19679} a_{19680} a_{19681} a_{19682} = 10000000020100020 11$

01211202121000001200211202020020020000201100221 2

0110221121L 2200201202020000

An example of a 6 × 9 binary M-array by folding a segment of above pseudorandom sequence is:

[1] A field with q elements is called a *Galois field* and is denoted by *GF(q)*.

$$\begin{bmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 0 & 1 & 2 & 2 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 & 0 & 0 \\ 1 & 1 & 0 & 2 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \tag{4}$$

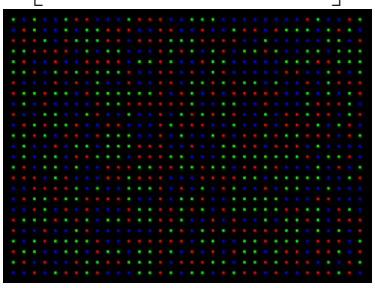


Fig. 1 Coloured spot pattern encoded according to an M-array.

The obtained pattern is shown in Fig. 1 where color blue has been matched with symbol 0, green with symbol 1 and red with symbol 2.

B. Pattern Segmentation and Decoding

When the pattern is projected on an unknown object, the camera provides an image of the pattern deformed according to the object shape. Firstly, it is necessary to segment the pattern in the image, i.e. identifying which part of the image contains the projected pattern. One of the key advantages of using coded light is that the image processing is greatly simplified. Usually, with an appropriate camera aperture it is possible to perceive only the projected pattern with the rest of the scene being removed.

The next step in our method involves extracting characteristic spots (or finding the visible colored spots). Once the center of gravity of every spot is located and its color is identified, the decoding process can start.

Prevailing methods uses fixed size window conducting iterative and exhaustive search. This is the red window shown in Fig. 2. The brute force algorithm is quite time consuming. It did not take into account surface normal changes from image shading. And it also does not account for color spots location change in images taken from different angles. This can lead to cases of more than 9 spots in the window or less than 9 spots in the search window causing decoding errors.

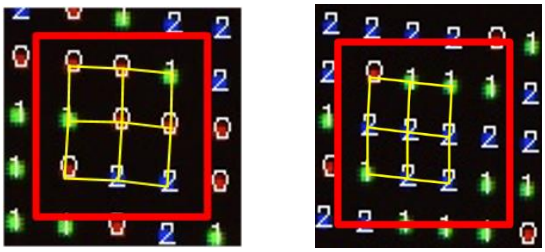
In view of that, the proposed method builds dynamic local window based on uniquely located color spots. This is shown as yellow window in Fig 2. As such, the dynamic window is generally not rectangular.

The proposed pattern decoding algorithm is shown in Algorithm 1. Assume n color spots have been extracted, choose $q_0(x_0, y_0)$ as center to construct the local window. The threshold value T in step 2 in Algorithm 1 is calculated by taking average of three shortest distances from all calculated distances between $q_0(x_0, y_0)$ and $q_i(x_i, y_i)$, $(i = 1, 2, \ldots, n-1)$ as:

$$T = Avg\{Min\{d, 3\}\} \tag{5}$$

where $[C, I] = Min\{d, j\}$ return minimal j elements from array d and I store the locations. $Avg\{\bullet\}$ is a function to take average.

The next step is to find 4-neighbour based on vertical-horizontal distance $v(i)$ and threshold value T. This refers to four spots a_2, a_8, a_4 and a_6 surrounding spot a_5 in Fig. 3. This is calculated as follows:



(a) (b)

Fig. 2 Window search comparison. (a) Prevailing fixed window search method. (b) Potential issues (i.e. including more than 9 spots) with fixed window searching.

$$v(i) = |x_i - x_0| + |y_i - y_0|, \ (i = 1, 2, \ldots, n-1) \tag{6}$$

$$[C, I] = Min\{v, 4\} \tag{7}$$

$$q_0 \in a_5 \tag{8}$$

$$q_{I(i)} \in \begin{cases} a_2 & if \ |x_{I(i)} - x_0| < k_1 \cdot T \ \text{ and } \ k_2 \cdot T < y_0 - y_{I(i)} < k_3 \cdot T \\ a_8 & if \ |x_{I(i)} - x_0| < k_1 \cdot T \ \text{ and } \ k_2 \cdot T < y_{I(i)} - y_0 < k_3 \cdot T \\ a_4 & if \ |y_{I(i)} - y_0| < k_1 \cdot T \ \text{ and } \ k_2 \cdot T < x_0 - x_{I(i)} < k_3 \cdot T \\ a_6 & if \ |y_{I(i)} - y_0| < k_1 \cdot T \ \text{ and } \ k_2 \cdot T < x_{I(i)} - x_0 < k_3 \cdot T \end{cases} \tag{9}$$

$$(i = 1, 2, 3, 4)$$

where k_1, k_2 and k_3 are weighting factor decided by user. A good try for these three can be 0.3, 0.7 and 1.3 respectively.

If 4-neighbourhood can be found, the next step is to find remaining 4-neighbour in the 3×3 local window. That is to find a_1, a_3, a_7 and a_9 in Fig. 4. This can be done with the help by drawing parallel lines. To illustrate, draw a line l_1 through a_4 parallel to line a_2a_5 and draw a line l_2 through a_2 parallel to line a_4a_5. And two lines l_1 and l_2 intersect at p. Neighbour point a_1 can be found by identifying the spot with mini-

Algorithm 1 Decoding based on dynamic local window.

Input: n identified color spots
1: **for** i = 0 to n-1 **do**
2: Input spot q_i (x_i, y_i) as center and calculate threshold T
3: **if** Search and locate 4-neighbourhood( )=false **Then** exit the loop
4: Construct local window
5: **if** Search and locate 8-neighbourhood( )=false **Then** exit the loop
6: **if** Spot decoding( )=true **Then** save decoded spot in the M-array
7: **end for**
8: Decoding consistency checking
9: Complementary identification
Output: decoded pattern

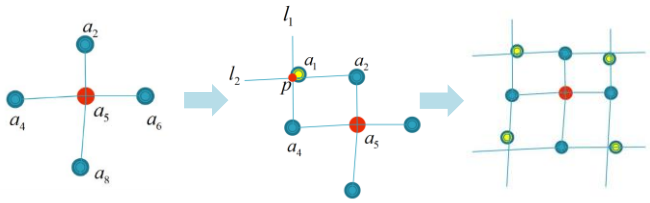


Fig. 3 Local window construction based on dynamic threshold value.



(a)

a_1	a_2	a_3
a_4	a_5	a_6
a_7	a_8	a_9

(b)

0	0	1
1	0	0
0	2	2

(c)

Fig. 4 Combination of code value in local window. (a) Local image with a 3×3 window. (b) Symbol of 9 spots in regular window. (c) Pattern window.

mum distance to point p. This distance should be within weighted threshold ($k_1 \cdot T$).

If 8-neighbourhood can be identified and local window can be constructed, the 9 element sequence can be extracted. This concludes step 5 in Algorithm 1.

Next step involves spot decoding. This is achieved by looking up the 9 element sequence in the M-array. In order to speed up this search, a look up table storing all the 3×3 windows included in the pattern is used. The decoded spot is saved in the M-array. This is step 6 in Algorithm 1. All n identified color spots will go through this loop (step 1-7). Most spots can be identified by the 9 windows of 3×3 in which it takes part. Those spots for which all the windows do not provide the same identification are removed.

Note that the decoding process is quite exigent and does not allow inconsistences or uncertainties. This can provoke an important number of spots to be rejected.

C. Decoding for Incomplete Windows

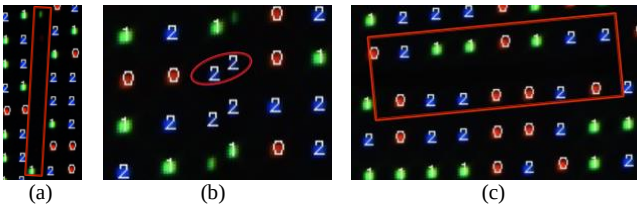
During the decoding, there are occasions of incomplete windows (i.e. more than 9 spots or less than 9 spots are included in the local window). This can be caused by various factors including occlusion, pattern breaking up, and deep surface discontinuity demonstrated in Fig. 5.

(1) Boundary spots decoding

For boundary spots, the 3×3 window cannot be built as shown in Fig. 6b with highlighted red window. However, it can be identified through yellow window with interior spot as center to construct the window. Red spots are boundary spots to be decoded. This involves building yellow windows to decode the boundary spots, which can be decoded 1-3 times depending on the point location.

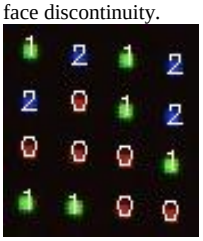
(2) Occlusion and deep surface discontinuity

For occlusion, this can be solved by applying image decoding from other images captured by different cameras. But

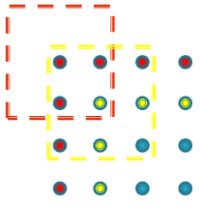


(a) (b) (c)

Fig. 5 Incomplete local windows. (a) Incomplete local window caused by occlusion. (b) Incomplete local window due to broken pattern. (c) Deep surface discontinuity.



(a)

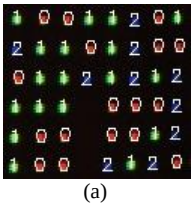


(b)

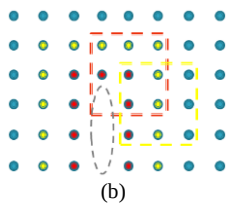
1	2	1	2
2	0	1	2
0	0	0	1
1	1	0	0

(c)

Fig. 6 Boundary spots decoding. (a) Local image. (b) Schematic diagram of spots. (c) Identified coding information.



(a)

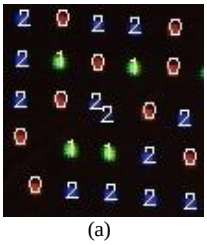


(b)

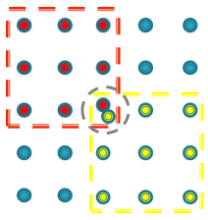
1	0	0	1	1	2	0	1
2	1	1	0	1	2	0	0
0	1	1	2	1	2	1	2
1	1	1		0	0	0	2
1	0	0		0	0	1	2
1	0	0		2	1	2	0

(c)

Fig. 7 Spots decoding with occlusion and deep surface discontinuity. (a) Local image. (b) Schematic diagram of spots. (c) Identified coding information.



(a)



(b)

2	0	2	2	0
2	1	0	1	0
2	0	X	0	2
0	1	1	2	0
0	2	2	2	2

(c)

Fig. 8 Spots decoding with broken pattern. (a) Local image. (b) Schematic diagram of spots. (c) Identified coding information.

in this way, the overall decoding accuracy will be discounted. As shown in Fig. 7b, the spots in gray dashed frame are occluded by foreground object in this camera perspective, which can be identified in other camera perspective. In addition, the captured spots can be treated similar to boundary spots and can be identified through yellow window with interior spot as center to construct the window. Based on the dynamic window which takes into account local information, the correspondence issue for image decoding on spots with deep surface discontinuity can be tackled effectively.

(3) Broken pattern

For broken pattern, shown with the gray dashed circle in Fig. 8b, some spots suffering from projection are cut in half by

the outline of object and can be identified through different windows with other spots as center such as red and yellow window. Therefore, the area will eventually be stored in two different spots, as a result, the coding information of the area be rejected.

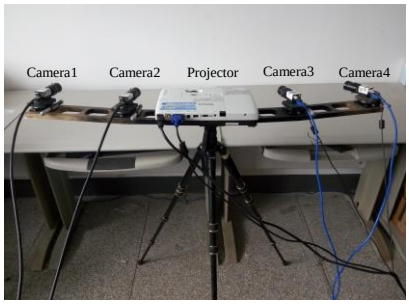
IV. Experiments

The vision system in the laboratory for the experiment consists of one projector, 4 cameras, one computer and one platform shown in Fig. 9. The projector (EPSON, EB-C301MS) is provided with 3000Lm brightness, 2000:1 contrast and 0.67–8.18m working distance. It is connected to the computer (3.4GHz CPU and 8GB RAM), which is controlled to generate the color encoded pattern and used for other techniques and algorithms, such as calibration, image processing and 3D reconstruction computation. The CCD cameras (2 FL2G-13S2C-C, 2 BFLY-PGE-13S2C-CS) have 1/3 inch sensor and 6mm lens.

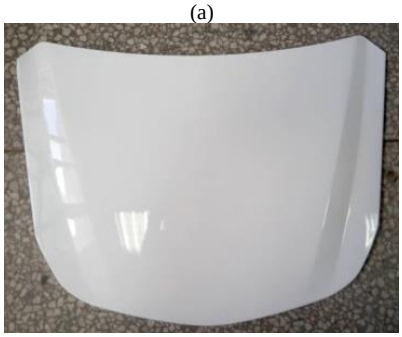
The experiment is conducted on a smooth and reflective car bonnet (or hood) shown in Fig. 9b. An encoded pattern of 25×34 is generated using three-color (red, blue and yellow) set, and is used to illustrate (through the Projector in Fig. 9a) the car bonnet model.

TABLE I
Results of Pattern Decoding

Camera Number	Failed	Wrong	Decoding success rate
Camera 1	17	0	98%
Camera 2	3	0	99.65%
Camera 3	3	0	99.65%
Camera 4	5	0	99.41%

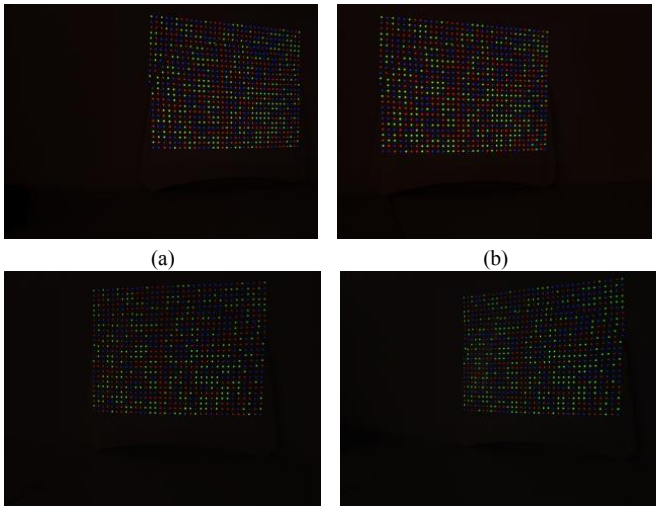


(a)



(b)

Fig. 9 Stereoscopic vision system involved in the experiment. (a) The setup. (b) A car bonnet (or hood) involved in the experiment.



(a) (b) (c) (d)

Fig. 10 Car bonnet (or hood) model and images captured from different cameras. (a-d) Images captured from Camera 1, Camera 2, Camera 3 and Camera 4, respectively.

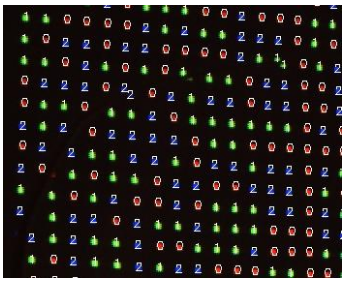


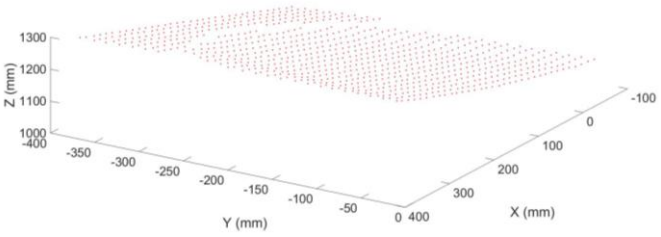
Fig. 11 Decoded pattern using proposed method.

The projected pattern is imaged by the four cameras with results shown in Fig. 10. Due to different camera shooting angles, there are variations in the arrangement of feature spots, especially for the two outbound cameras. Applying the proposed dynamic window method and developed efficient algorithm, the decoded results is shown in Fig. 11.

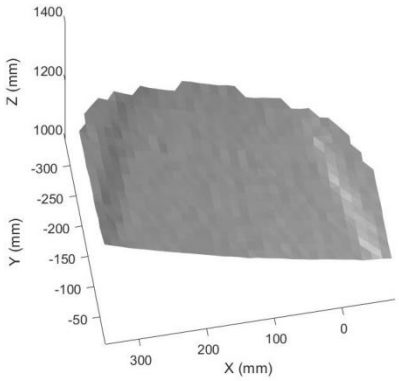
Results in Table I show that there are no decoding errors occurred in this experiment. For example there's no mis-pairing (mis-matching) in applying the developed method in this case. However, due to segmentation of feature spot at the outline of the car bonnet in the images captured by Camera 2 and Camera 3, there are 3 feature spots decoding failures.

The shooting angle of Camera 1 and Camera 4 leads to occlusion, resulting in 17 and 5 point decoding failures, respectively. The decoding success rate is quite high (98%~99.65%) compared with reported (90%-93%) on planar object and (37%~40%) on non-planar object using Morano's method [27].

The reconstructed car bonnet, using proposed decoding method, is shown in Fig. 12. The outlines of the reconstructed car bonnet from point clouds (Fig. 12a) are clear. The reconstructed car hood (Fig. 12b) is quite realistic. Even detailed curvature changes can be viewed clearly. Such good recon-



(a)



(b)

Fig. 12 Reconstructed car hood surface. (a) Reconstruction based on point clouds. (b) Reconstructed car hood shown in 3D surface.

struction result is due to high decoding success rate achieved by the proposed method.

V. Conclusion and Future Works

In this paper, a novel method for M-array image decoding in multi-view object shape reconstruction is presented. We focus on the order arrangement of the spots in the image captured by cameras, which is usually disorganized due to the non-planar surface geometry and occlusion induced discontinuity. Compared with existing adopted fixed rectangular window, the window is dynamic under proposed method and it can take into account local information during the image decoding. This substantially improves successful decoding rate, which has critical impact on the resolution of final reconstructed shape. The decoding process is speedy in comparison with existing brute-force algorithm approach. This will have big impact on many real-time applications where fast decoding is critical.

Built on the theory of multi-view stereoscopic vision and spatial coding method, the proposed methodology has solid research foundation. And it overcomes the difficulties of thin textures (or textureless) in surface reconstruction. The developed technique facilitates fast, high quality reconstruction of objects with small memory footprint and lower power consumption and at higher resolution than existing approaches.

Experiments conducted in laboratory show good results. However, further work is needed on exploring more flexible pattern configuration mode (size, primitive and color) in order to handle objects of different size and point clouds of different densities.

References

[1] C. Steger, M. Ulrich and C. Wiedemann. *Machine vision algorithms and applications*, John Wiley & Sons, 2018.

[2] K. Sjöö, A. Aydemir and P. Jensfelt, "Topological spatial relations for active visual search," *Robotics & Autonomous Systems*, vol. 60, pp. 1093-1107, 2012.

[3] J. Hafeez, S. C. Kwon, S. H. Lee and A. Hamacher, "3D surface reconstruction of smooth and textureless objects," in *International Conference on Emerging Trends & Innovation in ICT*, 2017, pp. 145-149.

[4] B. Yang and S. Li, "Pixel-level image fusion with simultaneous orthogonal matching pursuit," *Information Fusion*, vol. 13, pp. 10-19, 2012.

[5] I. Ishii, K. Yamamoto, K. Doi and T. Tsuji, "high-speed 3D image acquisition using coded structures light projection," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 925-930, 2007.

[6] D. Caspi, N. Kiryati, and J. Shamir, "Range imaging with adaptive color structured light," *IEEE Transactions on Pattern Analysis & Machine Intelligence*, vol. 20, pp. 470-480, 1998.

[7] D. Hung, "3D scene modelling by sinusoid encoded illumination," *Image and Vision Computing*, vol. 11, pp. 251-256, 1993.

[8] L.Zhang, B. Curless and S.M. Seitz, "Rapid shape acquisition using color structured light and multi-pass dynamic programming," in *Int. Symposium on 3D Data Processing Visualization and Transaction*. *Padova. Italy*, pp. 24-36, 2002.

[9] J. L. Posdamer and M. D. Altschuler, "Surface measurement by space-encoded projected beam systems," *Computer Graphics & Image Processing*, vol. 18, pp. 1-17, 1987.

[10] S. Inokuchi, K. Sato, and F. Matsuda, "Range imaging system for 3-d object recognition," in *Proc. Int. Conf. Pattern Recognit.*, 1984, pp. 1443-1454.

[11] R. Porras-Aguilar and K. Falaggis, "Absolute phase recovery in structured light illumination systems: Sinusoidal vs. intensity discrete patterns," *Optics & Lasers in Engineering*, vol. 84, pp. 111-119, 2016.

[12] T. Pribanić, S. Mrvoš, and J. Salvi, "Efficient multiple phase shift patterns for dense 3D acquisition in structured light scanning," *Image & Vision Computing*, vol. 28, pp. 1255-1266, 2010.

[13] E. Lilienblum and B. Michaelis, "Optical 3D Surface Reconstruction by a Multi-Period Phase Shift Method," *Journal of Computers*, vol. 2, pp. 73-83, 2007.

[14] D. Bergmann, "New approach for automatic surface reconstruction with coded light," *Proceedings of SPIE - The International Society for Optical Engineering*, vol. 2572, pp. 2-9, 1995.

[15] W. Yin, H. Cui, C. Zhang, X. Cheng, and D. Li, "A colorful codification method for structured light measurement based on the hue of single image," in *Optical Metrology and Inspection for Industrial Applications IV*, 2016, p. 100231N.

[16] M. Ribo and M. Brandner, "State of the art on vision-based structured light systems for 3D measurements," in *International Workshop on Robotic Sensors: Robotic and Sensor Environments*, 2005, pp. 2-6.

[17] S. Y. Zheng, H. L. Hu, X. Xu, and K. Le, "Structured-light-based 3D Reconstruction Using Gray Code and Line-shift Stripes," in *Progress in Measurement and Testing—proceedings of 2010 International Conference on Advanced Measurement and Test*, 2010, pp. 799-804.

[18] J. Salvi, J. Pagès, and J. Batlle, "Pattern codification strategies in structured light systems," *Pattern Recognition*, vol. 37, pp. 827-849, 2004.

[19] R. Benveniste and C. Ünsalan, "Nary coded structured light-based range scanners using color invariants," *Journal of Real-Time Image Processing*, vol. 9, pp. 359-377, 2014.

[20] P. Vuylsteke and A. Oosterlinck, "Range Image Acquisition with a Single Binary-Encoded Light Pattern," *IEEE Transactions on Pattern Analysis & Machine Intelligence*, vol. 12, pp. 148-164, 1990.

[21] T. Etzion, "Constructions for perfect maps and pseudorandom arrays," *IEEE Transactions on Information Theory*, vol. 34, pp. 1308-1316, 1988.

[22] C. Albitar, P. Graebling, and C. Doignon, "Design of a Monochromatic Pattern for a Robust Structured Light Coding," in *IEEE International Conference on Image Processing*, 2007, pp. VI - 529-VI - 532.

[23] H. Morita, K. Yajima and S. Sakata, “Reconstruction of surfaces of 3d objects by m-array pattern projection method,” in *Second International Conference on Computer Vision*, pp. 468-473, 1998.
[24] R. A. Morano, C. Ozturk, R. Conn, S. Dubin, S. Zietz, and J. Nissanov, “Structured Light Using Pseudorandom Codes,” *Pattern Analysis & Machine Intelligence IEEE Transactions on*, vol. 20, pp. 322-327, 1998.
[25] F. J. Macwilliams and N. J. A. Sloane, “Pseudo-random sequences and arrays,” *Proceedings of the IEEE*, vol. 64, pp. 1715-1729, 1976.
[26] T. Etzion, *Constructions for perfect maps and pseudorandom arrays*, Piscataway, N.J.: IEEE Press, 1988.
[27] J. Pages, C. Collewet, F. Chaumette, J. Salvi, S. Girona and F. Rennes, “An approach to visual servoing based on coded light,” *IEEE International Conference on Robotics and Automation, ICRA*, vol. 6, pp. 4118-4123, 2006.